

A Project Report on Skill Exchange System

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SUPERVISOR RECOMENDATION

This is to certify that this project entitled, “**Skill Exchange System**” prepared and submitted by **Anubhav Dahal, Sulav Raj Shrestha, Grishma Adhikari** and **Krish Shrestha** in partial fulfillment of the requirements of the degree of Bachelor of Computer System and Information Technology (BCSIT) awarded by Pokhara University, has been completed under my supervision. I recommend the same for acceptance by the University.

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CERTIFICATE OF APPROVAL

This project entitled, “**Skill Exchange System**” prepared and submitted **Anubhav Dahal, Sulav Raj Shrestha, Grishma Adhikari** and **Krish Shrestha**, has been examined and is accepted for the award of the degree of Bachelor of Computer System and Information Technology (BCSIT) awarded by Pokhara University.

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ABSTRACT

Skill Exchange System which is called SkillSwap, is a web-based platform designed to connect individuals who require services with skilled individuals who can provide them. In many communities, people face difficulties in finding reliable service providers for tasks such as tutoring, technical support, repairs, and other everyday services. At the same time, many skilled individuals lack a proper platform to showcase their abilities and reach potential customers.

The main objective of this project is to develop a centralized and user-friendly system that facilitates skill sharing and service exchange within the community. The platform allows users to register, create profiles, search for available services, and connect with skilled workers based on their requirements. The system also supports service management, booking requests, and secure user authentication.

The Skill Swap platform was developed using HTML, CSS, JavaScript, Node.js, Express.js, and MySQL. Various software engineering methodologies were applied during requirement analysis, system design, implementation, and testing phases to ensure the reliability and usability of the system.

The proposed system aims to improve accessibility to local services, promote community collaboration, and provide opportunities for skilled individuals to gain recognition and employment. Future enhancements include complete user profile management, session booking, ratings and reviews, real-time communication, online payment integration, and mobile application support.

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CHAPTER 1: INTRODUCTION

1.1. Background of the Study

With the rapid growth of digital technology, online platforms have become an effective medium for connecting individuals and providing services. Many people possess valuable skills and expertise that remain underutilized, while others often struggle to find reliable assistance for small tasks and services. Traditional methods of finding skilled individuals are often time-consuming, expensive, and lack transparency.

Skill Swap (SS) is a web-based application developed to bridge the gap between people who need assistance and those who can offer their skills. The system provides a centralized platform where users can showcase their abilities, search for services, communicate with others, and request assistance. The platform aims to promote community-based skill sharing by enabling direct interaction between service seekers and service providers.

The project is based on the concepts of the sharing economy and peer-to-peer service exchange, where users can connect directly without relying on intermediaries. By creating a trusted environment with user profiles, ratings, reviews, and communication features, Skill Swap encourages collaboration and efficient utilization of community skills.

1.2. Statement of the Problem

Many individuals face difficulties in finding trustworthy and skilled people for small tasks such as tutoring, repairs, technical assistance, and other everyday services. Existing solutions often focus on paid services or online learning platforms and provide limited support for local community-based interactions.

The major problems identified are:

- i. **Lack of a Centralized Platform:** There is an absence of a unified, localized digital directory dedicated to immediate, task-oriented skill exchanges, forcing individuals to rely on fragmented, inefficient communication channels like social media groups or word-of-mouth.
- ii. **Information Asymmetry and Trust Deficit:** Engaging unknown independent service providers introduces significant risks regarding reliability, quality, and personal safety. Without a transparent, historical verification system, community trust remains compromised.
- iii. **Economic Inefficiencies:** For minor, non-professional tasks, mainstream commercial service providers charge premium rates driven by administrative overheads and middleman interventions, making simple tasks financially burdensome for demographic segments like university students.
- iv. **Underutilization of Local Talent:** Competencies possessed by community members frequently remain dormant or unmonitored, limiting opportunities for experiential learning, peer-to-peer mentorship, and mutual community support.

Therefore, there is a need for a system that can efficiently connect people, promote trust and transparency, and facilitate skill sharing within communities.

1.3. Objectives of the Study

The main objective of the project is to develop a web-based Skill Swap platform that connects individuals seeking assistance with those willing to offer their skills.

The specific objectives are:

i. To Connect Customers with Skilled Workers:

The project aims to create a connection between customers and skilled workers. Many people spend a lot of time searching for reliable workers through friends, advertisements, or social media. The proposed system allows users to quickly find workers based on their skills, location, and availability. Skilled individuals can create profiles to display their expertise and services, while customers can easily search and contact suitable workers. This improves job opportunities for workers and helps customers receive services more conveniently.

ii. To Promote Skill Sharing in the Community:

Another objective is to encourage skill sharing within the community. Many talented individuals do not have a proper platform to showcase their abilities. This system provides them with opportunities to offer their services and gain recognition. The platform also supports local workers by helping them earn income and improve their professional growth. As a result, communities become more self-dependent by accessing local services easily.

iii. To Make Service Booking Easy and Fast:

The project simplifies the traditional service booking process, which is often slow and inconvenient. Through the platform, users can browse services, compare workers, and book appointments quickly. Features such as service categories, search filters, booking forms, and notifications make the process faster and more user-friendly. This saves time for both customers and workers while improving overall convenience.

iv. To Build Trust Through Ratings and Reviews:

The system includes ratings and reviews to increase trust between customers and workers. After completing a service, customers can provide feedback based on their experience. Positive reviews help workers build a good reputation, while feedback encourages better service quality. This feature

increases transparency and helps users make informed decisions before booking services.

v. To Manage Offline Services Efficiently:

Although the platform is digital, most services are provided offline. Therefore, the system helps manage appointments, service requests, and communication between customers and workers effectively. Once a booking is confirmed, both parties receive important details such as date, time, and location. The platform also stores service records for future references, making the process more organized and reliable.

1.4. Scope of the Study

The primary focus of the Skill Swap system is to design, develop, and implement a secure, functional, and community-centric web platform that facilitates peer-to-peer skill matching and micro-task collaboration. The platform bridges the gap between individuals who possess specific practical or academic competencies and those who require immediate assistance with localized tasks.

To ensure structured implementation, the operational boundaries and functional scope of this project are explicitly defined as follows:

- i. **User Registration and Authentication Module:** Implementation of secure onboarding workflows that allow new users to register an account. It includes field validation, unique email constraints, and a secure login/logout mechanism powered by encrypted session management to protect user data identity.
- ii. **Dynamic Profile Management:** Provision of a personalized user workspace where individuals can update their contact records, upload short bio descriptions, manage their availability statuses, and alternate fluidly between acting as a Service Seeker or a Service Provider.
- iii. **Skill Cataloging and Repository Management:** Tools for Service Providers to list their unique competencies. This includes specifying descriptive titles, detailed task explanations, taxonomy classifications (e.g., Tutoring, Home Repairs, Tech Support), and hourly or task-based rate estimations.
- iv. **Search and Discovery Engine:** Development of an interactive, client-side exploration grid. Users can look up services using text-based queries and apply specific category filters to instantly discover available skill assets within their community.
- v. **Booking and Reservation Workflow:** A systematic state-tracking pipeline that manages service requests. It handles everything from initial booking creation by a seeker, to pending notifications, and provider action controls to accept, reschedule, or complete a task.
- vi. **Internal Communication Channel:** Integration of a message routing system within the web application. This allows matched users to securely discuss task details, negotiate terms, and coordinate logistics prior to meeting.

- vii. **Reputation Management System:** Implementation of a post-service feedback loop. Users can submit numerical star ratings (1 to 5) and textual reviews, which the system uses to dynamically calculate a provider's public trustworthiness index.
- viii. **Cross-Modal Exchange Handling:** System support for tracking the operational lifecycles of both virtual online sessions (such as remote digital tutoring) and physical, neighborhood-level offline exchanges (such as hands-on appliance repairs).

1.5. Limitations of the Study

While the Skill Swap platform offers a robust and practical framework for localized service matching, certain technical, environmental, and operational boundaries constrain its current deployment version:

- i. **Platform Dependency:** The current system architecture is developed exclusively as a responsive web-based application. While it functions smoothly across modern mobile web browsers, it lacks a dedicated, native mobile application (iOS/Android) wrapper.
- ii. **Identity and Information Verification Constraints:** The platform lacks integration with automated, third-party citizen verification APIs or official background check systems. Consequently, the accuracy, quality, and legitimacy of user-provided certificates and listed skills rely heavily on self-reported user honesty.
- iii. **Community Scaling Dependency:** The marketplace logic and overall utility of the platform are strictly dependent on critical mass. The system requires consistent, active participation from a balanced ratio of local skill providers and seekers to prevent data stagnation.
- iv. **Absence of Real-Time Location Tracking:** The application relies on static, user-declared location strings or general campus/neighborhood regions. It does not utilize real-time GPS tracking or automated geospatial distance algorithms to calculate proximity matrices.
- v. **Moderation and Quality Assurance Overhead:** The system provides the foundational architecture for feedback but lacks automated content filtering or automated dispute-resolution mechanisms. This means manual administrative review is still required to handle fraudulent listings or platform abuse.
- vi. **Network Infrastructure Reliance:** Because the application relies on an active client-server architecture using a remote MySQL and Node.js backend, users must have continuous internet connectivity to access dashboards, list skills, or update booking states. Offline functional access is not supported.

CHAPTER 2: REVIEW OF LITERATURE

2.1. Description of Fundamental Theories, General Concepts and Terminologies

The Skill Swap platform is developed based on the concepts of the Sharing Economy and Peer-to-Peer (P2P) Service Exchange Systems. The sharing economy refers to an economic model where individuals share resources, skills, and services directly with one another through digital platforms rather than relying on traditional organizations or intermediaries. This approach promotes efficient utilization of available skills within a community while reducing costs and increasing accessibility to services.

Similarly, the peer-to-peer model enables direct interaction between users who need services and users who are willing to provide them. Instead of depending on third-party agencies, individuals can communicate, exchange information, and arrange service requests through a centralized online platform. This concept forms the core foundation of the Skill Swap system.

The platform incorporates several important concepts and terminologies that are essential for its operation:

- User Roles:

The system allows users to perform multiple roles based on their needs. A user can act as a Service Seeker, who requests assistance for a particular task, or as a Service Provider, who offers a skill or service to others. This flexibility encourages active participation and promotes mutual skill sharing within the community.

- Session Management:

Session management is a backend mechanism used to maintain and monitor authenticated user activities throughout their interaction with the system. Once a user logs into the platform, a secure session is created that allows the user to access protected features without repeatedly entering login credentials. This improves both security and user experience.

- **Booking and Request Management System:**

The booking and request management system handles the complete workflow of service exchanges. Users can submit requests for services, which are then received by service providers. Providers can accept, reject, or manage requests according to their availability. The system tracks the status of each request from creation to completion, ensuring proper record keeping and transparency.

- **Communication System:**

Effective communication is a key component of the Skill Swap platform. The system provides messaging and interaction features that enable seekers and providers to discuss service details, schedules, expectations, and other relevant information before confirming a service exchange. This helps reduce misunderstandings and improves coordination between users.

- **Rating and Review System:**

To establish trust and credibility within the platform, a rating and review mechanism is incorporated. After completing a service exchange, users can provide ratings and feedback based on their experience. These ratings help future users evaluate the reliability and quality of service providers, creating a transparent and trustworthy environment.

- **Community-Based Service Exchange:**

Unlike many existing platforms that focus primarily on paid services or online learning, Skill Swap emphasizes community-driven collaboration. The platform encourages local interaction, skill utilization, and mutual support among community members, helping individuals connect with trusted people in their surrounding areas.

By integrating these concepts and technologies, the Skill Swap platform provides a reliable, efficient, and user-friendly environment for connecting skilled individuals with people seeking assistance, ultimately promoting collaboration and community development.

2.2. Review of the similar Projects and Theories

Several existing skill-sharing and online learning platforms were studied to understand their features, strengths, and limitations. Platforms such as Skillshare, Skillpact, and Skilljar provide online environments where users can learn new skills through courses, training materials, and collaborative activities.

These systems commonly offer features such as:

- User registration and authentication
- Skill or course listing
- Communication and collaboration tools
- Rating and review mechanisms
- Progress tracking and learning resources

Despite their usefulness, these platforms have certain limitations:

- They primarily focus on online learning and educational content.
- Opportunities for local community interaction are limited.
- Support for face-to-face skill exchange and offline collaboration is minimal.
- Real-world peer-to-peer service exchange is often unavailable.
- Many platforms emphasize paid courses rather than community-based knowledge sharing.

The SkillSwap system extends beyond traditional online learning platforms by creating a community-oriented environment where users can both learn and provide skills. Unlike existing systems that mainly focus on educational courses, SkillSwap supports a wider range of skills and services such as technology, design, language learning, tutoring, home maintenance, photography, cooking, arts, and other community-based activities.

The proposed system offers several advantages:

- Supports offline skill exchange which results in better efficiency.
- Encourages local community engagement and collaboration.

- Allows direct interaction between skill providers and skill seekers.
- Facilitates peer-to-peer service exchange rather than only course-based learning.
- Provides service booking and scheduling functionality.
- Creates opportunities for practical, real-world learning experiences.
- Promotes knowledge sharing within local communities.
- Enables users to build reputation through ratings and reviews.

Several research studies related to peer-to-peer learning and skill exchange were also reviewed during the requirement identification phase.

A study by Sri (2025) emphasized the importance of user profiles, communication systems, and collaborative learning in skill-sharing applications [1]. Similarly, Jadhav et al. (2025) highlighted matchmaking mechanisms and community interaction as key components of successful peer-to-peer skill exchange platforms [2].

Anam et al. (2025) proposed a web-based collaborative learning platform developed using PHP and MySQL, providing features such as authentication, user management, and skill administration [3]. Furthermore, Ukata and Moko (2026) discussed the effectiveness of peer-to-peer interaction and practical learning through mobile-based skill exchange applications [4].

These studies helped identify important system requirements such as communication, booking management, user profiles, and community-based interaction for the proposed system.

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

3.1. System Analysis

System analysis is an important phase of the Software Development Life Cycle (SDLC) that focuses on understanding the problems, requirements, and objectives of a system before development begins. It involves studying user needs, identifying system functionalities, and determining how the proposed system will solve existing problems.

For the Skill Swap system, system analysis was conducted to understand the requirements of service seekers, service providers, and administrators. This process helped identify the necessary features, data requirements, system processes, and operational constraints. It also included evaluating the feasibility of the system, analyzing data flow, and designing solutions that ensure efficient skill sharing and service exchange within the community.

3.1.1. Requirement Analysis

3.1.1.1.Function Requirements:

Functional requirements describe the specific operations and services that the Skill Swap system must perform. These requirements are represented in the Use Case Diagram, which illustrates the interactions between different actors and the system.



Figure 1: Usecase Diagram

i. User Registration and Authentication:

The system shall allow users to create accounts, log in securely, and manage their personal profiles. Authentication mechanisms shall ensure that only authorized users can access protected system features.

ii. Profile Management:

Users shall be able to update personal information, profile pictures, skill details, contact information, and account preferences.

iii. Skill Management:

Service Providers shall be able to add, edit, update, and remove skills offered through the platform. Providers may also specify service descriptions, pricing information, and availability.

iv. Search and Service Discovery:

Service Seekers shall be able to search and browse available skills and services using keywords and filtering options to find suitable providers.

v. Booking and Scheduling:

The system shall allow seekers to send booking requests and select preferred dates and times. Providers shall have the ability to accept, reject, or manage incoming requests.

vi. Communication System:

The platform shall provide messaging functionality that enables direct communication between service seekers and service providers before and after booking confirmation.

vii. Payment Management:

The system shall support secure payment processing between seekers and providers while maintaining records of completed transactions.

viii. Rating and Review System:

After completing a service, users shall be able to submit ratings and reviews that contribute to the trustworthiness and reputation of service providers.

ix. Report Management:

Users shall be able to report inappropriate behavior, fraudulent activities, or policy violations. Administrators shall review and resolve reported issues.

x. Dashboard Management:

The system shall provide dashboards that allow users to monitor bookings, requests, payments, ratings, and other relevant activities.

3.1.1.2.Non-Functional Requirements:

Non-functional requirements define the quality attributes, performance expectations, and operational constraints of the Skill Swap system.

i. Performance Requirements:

The system shall provide fast response times for user interactions such as login, searching, booking, and messaging. Most user requests should be processed within a few seconds under normal operating conditions.

ii. Security Requirements:

User passwords shall be securely stored using encryption and hashing techniques. Authentication and session management mechanisms shall protect user accounts from unauthorized access. Sensitive information such as payment data shall be handled securely.

iii. Reliability Requirements:

The system shall maintain stable operation and provide consistent access to services with minimal downtime. Data integrity shall be preserved during all transactions and system operations.

iv. Scalability Requirements:

The system shall support future growth in the number of users, skills, bookings, and transactions without significant performance degradation.

v. Usability Requirements:

The user interface shall be simple, intuitive, and easy to navigate. Users with basic computer knowledge should be able to access system features without extensive training.

vi. Compatibility Requirements:

The web application shall be accessible through modern web browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari. The interface shall adapt to desktop, tablet, and mobile devices.

vii. Maintainability Requirements:

The system shall be developed using a modular structure, allowing future updates, bug fixes, and feature enhancements to be implemented efficiently.

viii. Availability Requirements:

The platform shall be available to users whenever internet connectivity is available, ensuring continuous access to services and user data.

3.1.2. Feasibility Analysis

3.1.2.1. Technical Feasibility

Our system is technically feasible because it can be developed using commonly available web technologies.

The technologies planned for development include:

- Frontend: HTML, CSS, JavaScript
- Backend: PHP
- Database: MySQL

The required tools, frameworks, and development environments are accessible and suitable for a student-level project. The development team has basic knowledge of these technologies, making implementation achievable.

3.1.2.2. Operational Feasibility

The system is operationally feasible because it is designed to be simple and user friendly.

Users can easily:

- Register and create profiles
- Search for services
- Communicate with providers
- Book services
- Make payments and provide feedback

The platform supports practical real-world interaction and addresses the problem of finding trusted local services efficiently.

3.1.2.3. Financial Feasibility

The project is economically feasible because the development cost is relatively low. Most of the required technologies and tools are open-source and freely available. The project mainly requires:

- Internet access
- Development tools
- Basic hosting services

Since the system is developed as an academic project, no major financial investment is required. Additionally, the platform can generate revenue in the future through commission-based payments from service transactions.

3.1.2.4. Schedule Feasibility

Schedule feasibility evaluates whether the proposed system can be developed and implemented within the available time frame. The development of the Skill Swap system was planned according to the academic project schedule and divided into several phases, including requirement analysis, system design, database design, interface development, backend development, testing, and documentation.

The project was considered schedule feasible because the required technologies such as HTML, CSS, JavaScript, Node.js, and MySQL are familiar to the development team and the project scope is manageable within the allocated semester duration. Proper planning and task distribution among team members helped ensure that each development phase could be completed on time.

Therefore, the Skill Swap system can be successfully developed, tested, and documented within the given project schedule without requiring additional time or resources.

3.1.3. Data Modeling

Data modeling describes the data required by the system and the relationships between different entities.

Main Entities

- User
- Service
- Booking
- Payment
- Review
- Admin

Relationships

- One User can create many Services.
- One Service belongs to one Service Provider.
- One Service can have many Bookings.
- One Booking belongs to one Service Seeker.
- One Booking generates one Payment.
- One Booking can receive one Review.
- Admin manages users, services, bookings, and payments.

The system stores user information, service details, booking records, payment information, and reviews. These entities are connected through primary and foreign keys to ensure data integrity and consistency.

3.1.4. Process Modeling

Process modeling describes how data flows through the system and how users interact with the platform.

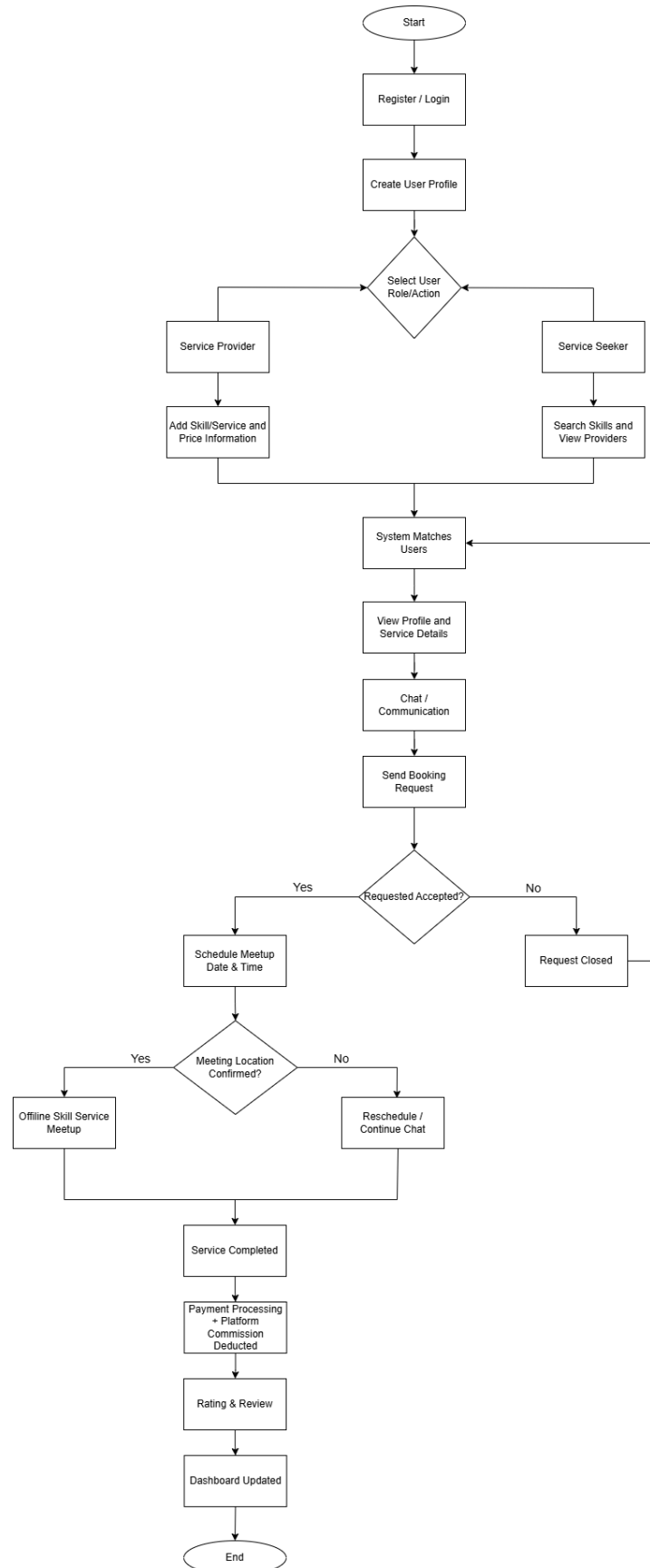


Figure 2: Flowchart

The system flowchart represents the overall workflow of the platform from user registration to service completion.

The process begins when users register and create profiles. Service providers add skills and pricing information, while service seekers search for services and providers. The system matches users based on their requirements and enables communication through the chat system.

After communication, booking requests are sent and managed by the provider. Once accepted, users schedule an offline meetup to complete the service. After successful completion, payment is processed, platform commission is deducted, and ratings and reviews are provided.

The flowchart helps visualize the step-by-step operation of the entire system.

3.2. System Design

The SkillSwap platform follows a three-tier architecture consisting of:

- Presentation Layer (Frontend)
- Application Layer (Backend)
- Data Layer (Database)

3.2.1. Architectural Design

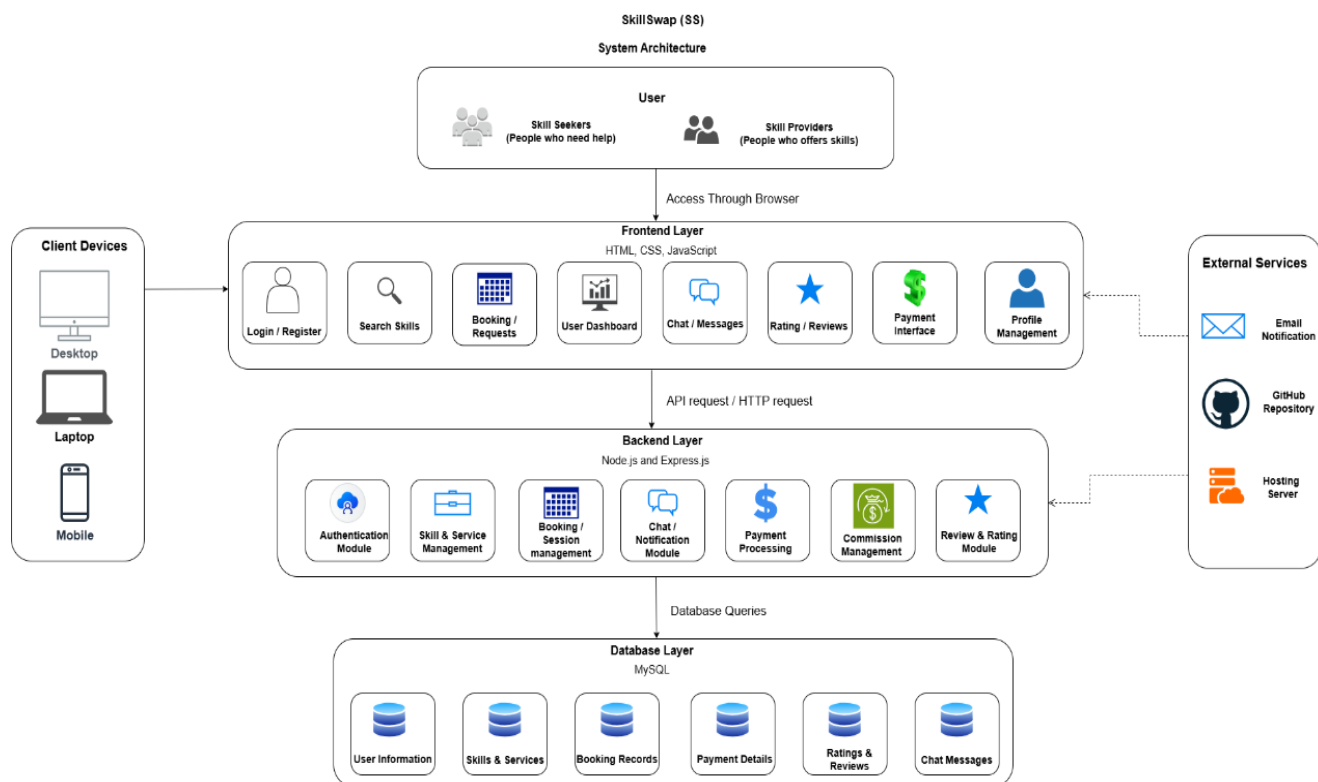


Figure 3: System Architecture

i. Presentation Layer (Frontend)

This layer handles user interaction and interface design. Users interact with the system through web pages developed using:

- HTML
- CSS
- JavaScript

This layer includes:

- Login pages
- Dashboard
- Skill listing pages
- Booking interfaces
- Chat interfaces

ii. Application Layer (Backend)

The backend processes business logic and system operations. It manages:

- User authentication
- Booking management
- Chat handling
- Payment processing
- Rating and review management

The backend will be developed using:

- Node.js
- Express.js

iii. Database Layer

This layer stores and manages all system data including:

- User information
- Skills and services
- Bookings
- Payments
- Ratings and reviews

The database system used is:

- MySQL

3.2.2. Database Schema Design

The database schema of the SkillSwap platform was designed using MySQL Workbench. The schema consists of several entities including User, Service, Booking, Payment, Review, and Admin. These entities are interconnected through primary and foreign key relationships to ensure data integrity and efficient data management.

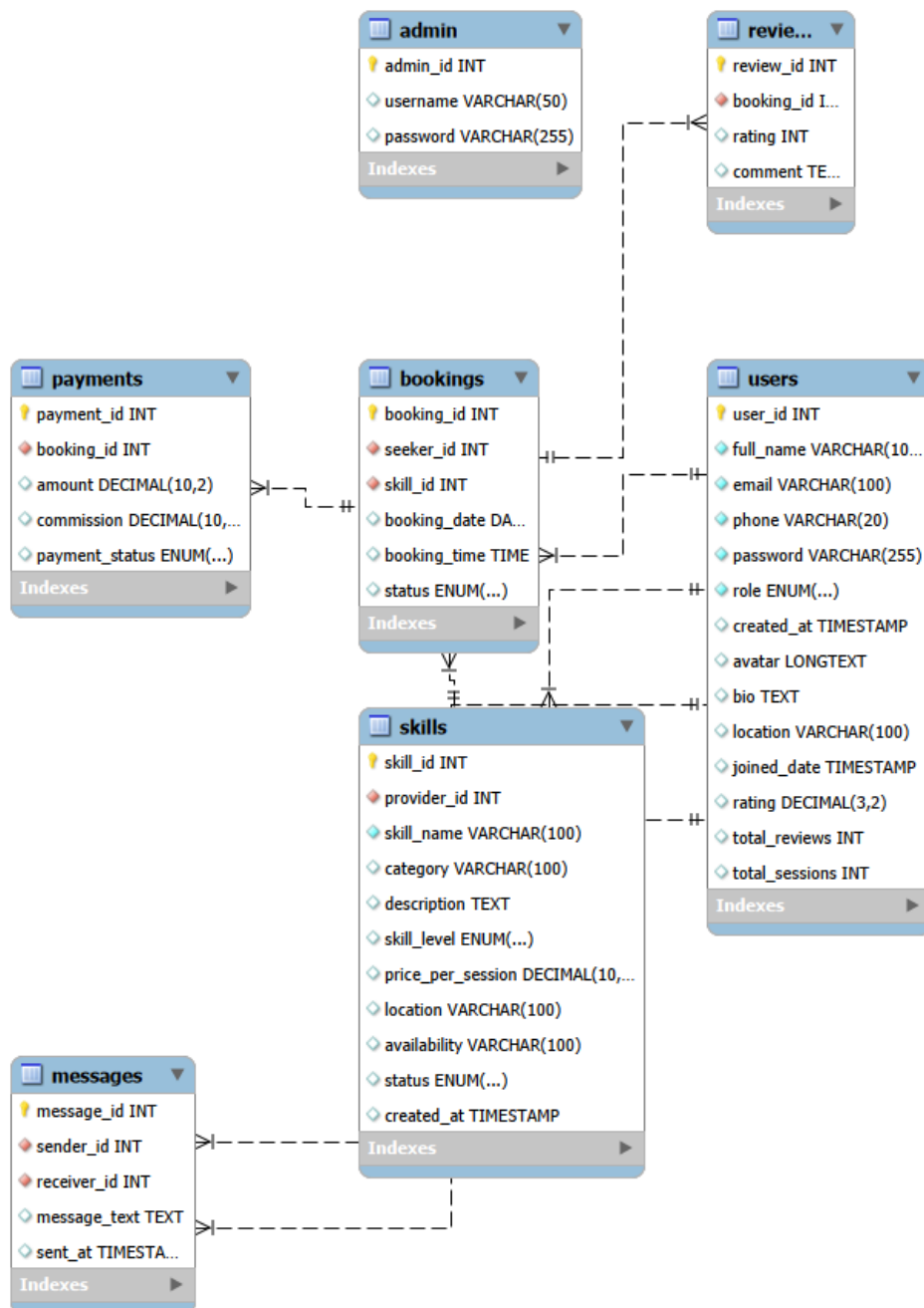


Figure 4: ER Diagram

3.2.3. Interface Design

3.2.4. Physical Model

Hardware Requirements

- Processor: Intel i3 or higher
- RAM: 4GB minimum
- Storage: 20GB free space
- Internet Connection

Software Requirements

- Windows 10/11
- VS Code
- Node.js
- MySQL Workbench
- Web Browser (Chrome, Firefox, Microsoft Edge)

CHAPTER 4: IMPLEMENTATION AND TESTING

4.1. Implementation

Implementation involves converting the system design into a working software application.

The SkillSwap platform was developed using Node.js, Express.js, MySQL, HTML, CSS, and JavaScript.

4.1.1. Tools Used

Table 1: Tools Used

Category	Tool
CASE TOOL	Figma
Programming Language	JavaScript
Frontend	HTML, CSS
Backend	Node.js, Express.js
Database	MySQL
IDE	VS Code
API Testing	Postman
Version Control	GitHub

4.1.2. Implementation Details of Modules

User Management Module

Functions:

- User Registration
- Login
- Logout
- Profile Management

Service Management Module

Functions:

- Create Service
- Edit Service
- Delete Service
- Search Services

Booking Module

Functions:

- Create Booking
- Cancel Booking
- View Booking Status

Payment Module

Functions:

- Process Payments
- Calculate Commission
- Store Transaction Records

Review Module

Functions:

- Submit Review
- View Reviews
- Rating Calculation

Admin Module

Functions:

- Manage Users
- Manage Services
- Monitor Payments
- Generate Reports

4.2. Testing

Testing was performed to verify that all modules function correctly and satisfy user requirements.

4.2.1. Test Cases for Unit Testing

Table 2: Test Cases for Unit Testing

Test Case ID	Module	Test Scenario	Expected Result	Status
UT-01	Registration	Enter valid user details	Account created successfully	Pass
UT-02	Registration	Empty full name	Error message displayed	Pass
UT-03	Registration	Invalid email	Validation error displayed	Pass
UT-04	Login	Correct email and password	User login successful	Pass
UT-05	Login	Incorrect password	Login denied	Pass
UT-06	Service Creation	Add service details	Service saved in database	Pass

4.2.2. Test Cases for System Testing

Table 3: Test Cases for System Testing

Test Case ID	System Function	Expected Result	Status
ST-01	User Registration → Login	User can login after registration	Pass
ST-02	Login → Home Page	Profile menu displayed after login	Pass
ST-03	Logout	User redirected to login page	Pass
ST-04	Unauthorized Access	User redirected to login page when accessing protected pages	Pass
ST-05	Database Connection	Data stored and retrieved correctly	Pass

CHAPTER 5: CONCLUSION AND FUTURE RECOMMENDATION

5.1. Lesson Learnt

The development of the Skill Swap platform provided valuable practical experience in software development and project management. Through this project, the team gained knowledge about requirement analysis, system design, database modeling, frontend and backend development, and software testing.

The project enhanced our understanding of web technologies such as HTML, CSS, JavaScript, Node.js, Express.js, and MySQL. We learned how different components of a web application interact to provide a complete and user-friendly system. The project also improved our teamwork, communication, problem-solving, and time-management skills.

Furthermore, we gained practical experience in applying software engineering concepts such as system analysis, database design, testing methodologies, and documentation. The project demonstrated the importance of understanding user requirements and designing solutions that address real-world problems effectively.

5.2. Conclusion

The Skill Swap platform was developed to provide a centralized environment where individuals can connect with skilled service providers and access various services conveniently. The system addresses the challenges of finding reliable assistance, promotes community-based skill sharing, and creates opportunities for individuals to showcase their talents and expertise.

The developed platform successfully implements core functionalities including user registration, authentication, profile management, service management, service discovery, and booking management. The system was tested to ensure that its major functions operate correctly and meet the specified requirements.

Overall, the project achieved its primary objective of creating a web-based platform that facilitates interaction between service seekers and service providers. The Skill

Swap platform has the potential to improve community collaboration, increase access to local services, and promote the effective utilization of available skills within society.

5.3. Recommendations

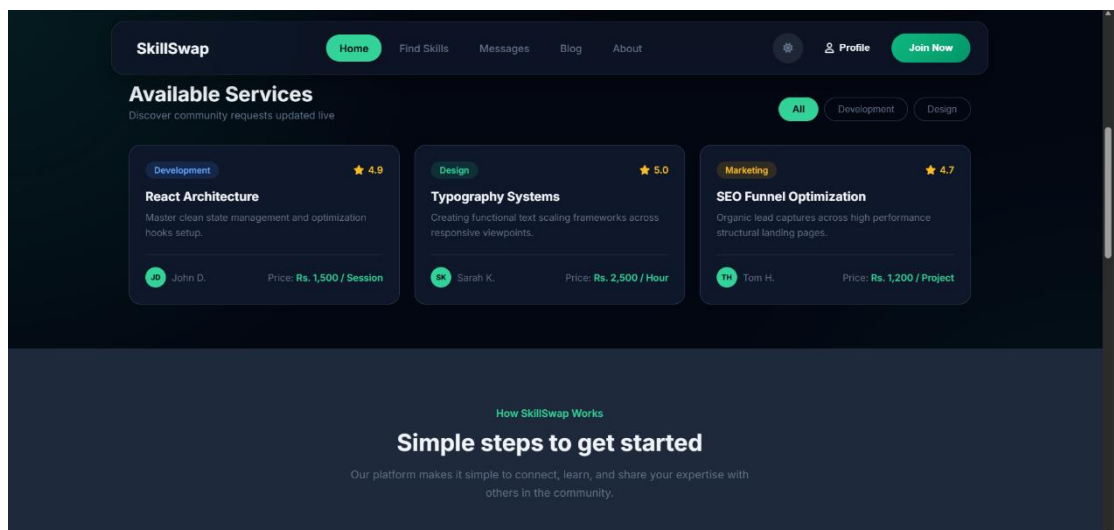
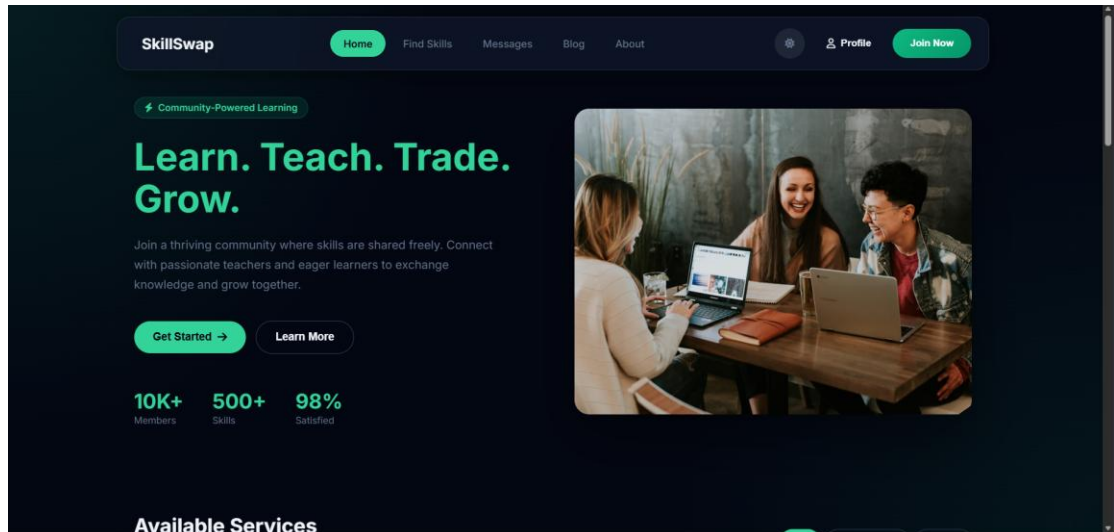
Although the current version of the Skill Swap platform provides essential functionality, several improvements can be implemented in future versions to enhance usability and overall system effectiveness.

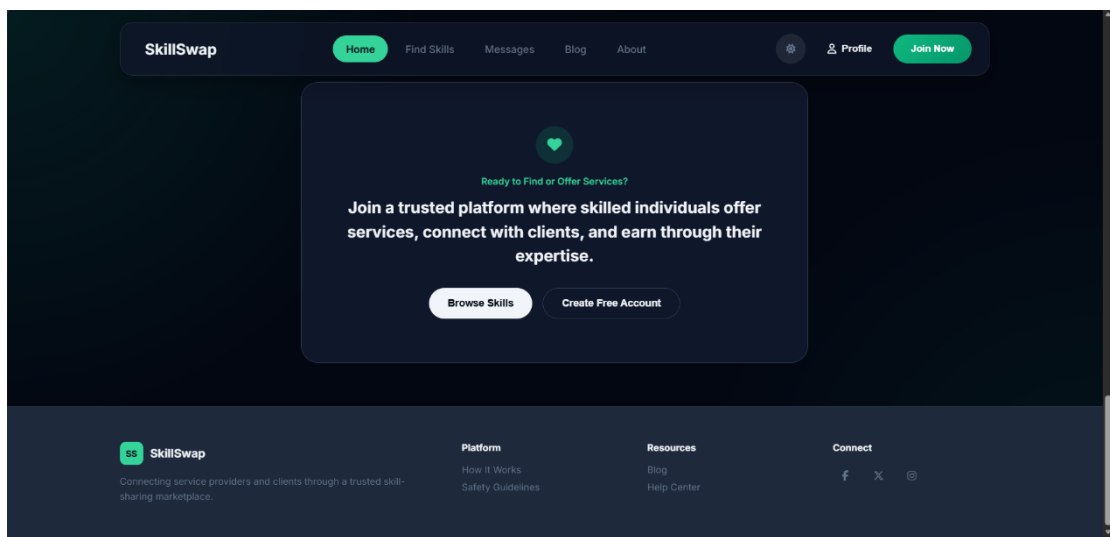
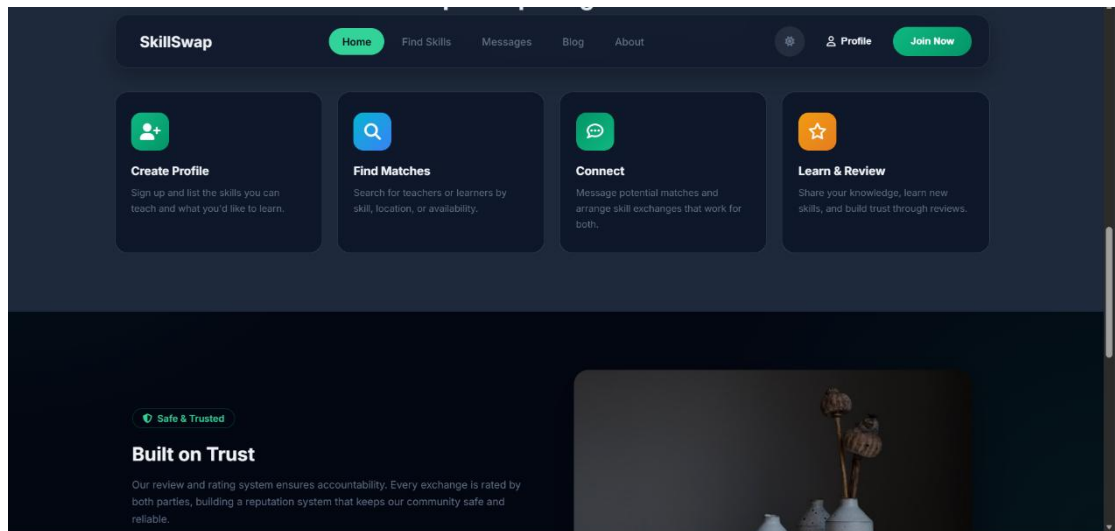
- i. Implement complete user profile management with certifications
- ii. Develop a full session booking and scheduling system with appointment management features.
- iii. Introduce ratings and reviews after service completion to improve trust and transparency.
- iv. Develop a dedicated mobile application for Android and iOS platforms.
- v. Strengthen security through user verification and improved authentication mechanisms.
- vi. Implement secure online payment gateway integration for service transactions.

APPENDICES

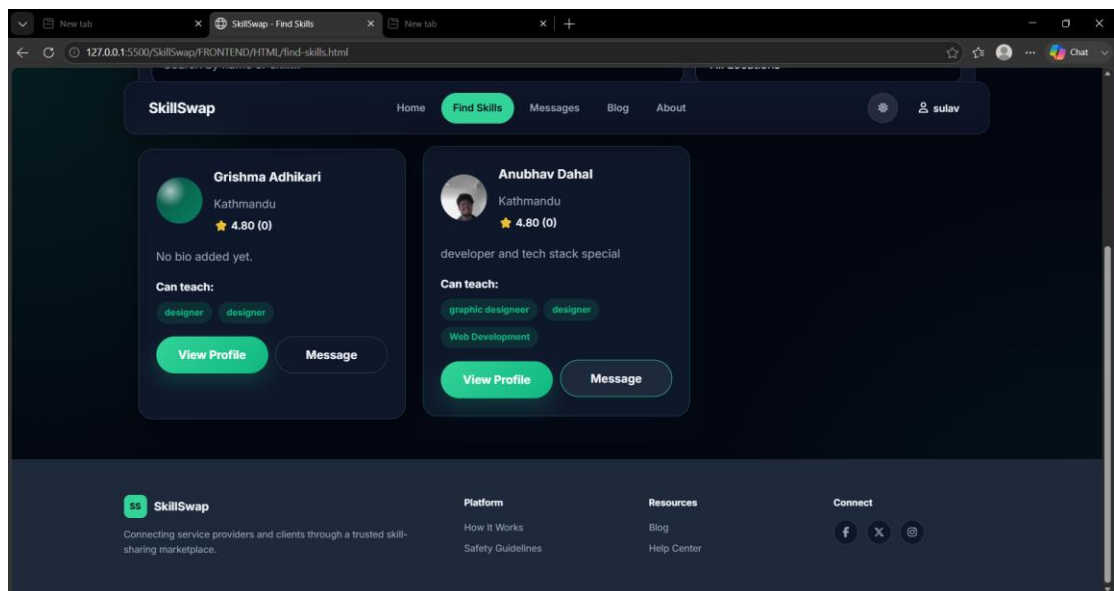
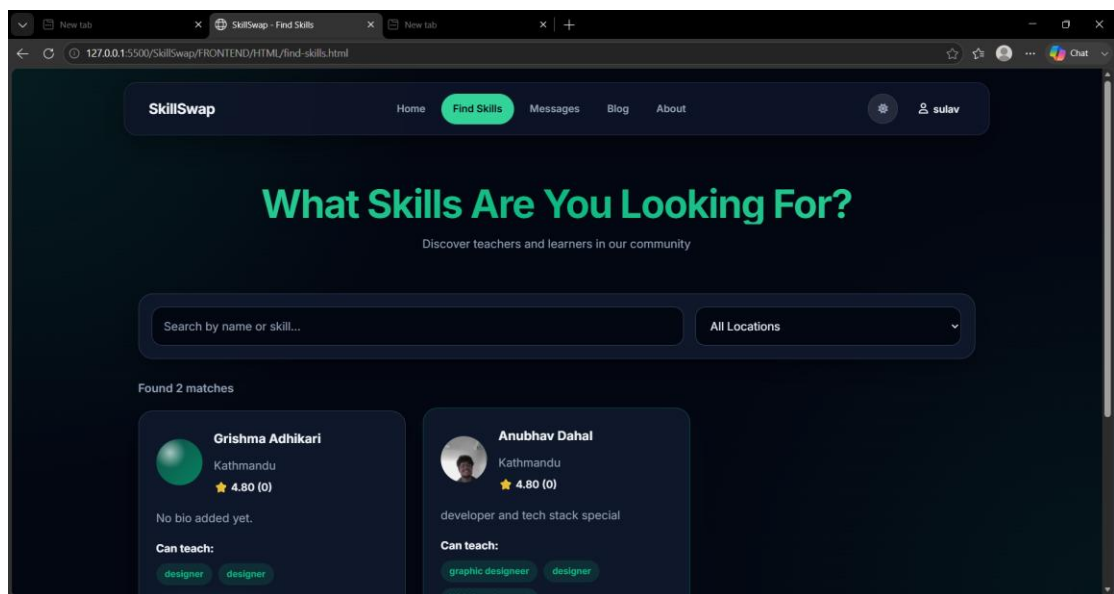
This appendix presents the screenshots of the major interfaces of the Skill Swap system. These interfaces demonstrate the visual design, navigation structure, and functionalities implemented in the application.

Home Page Interface

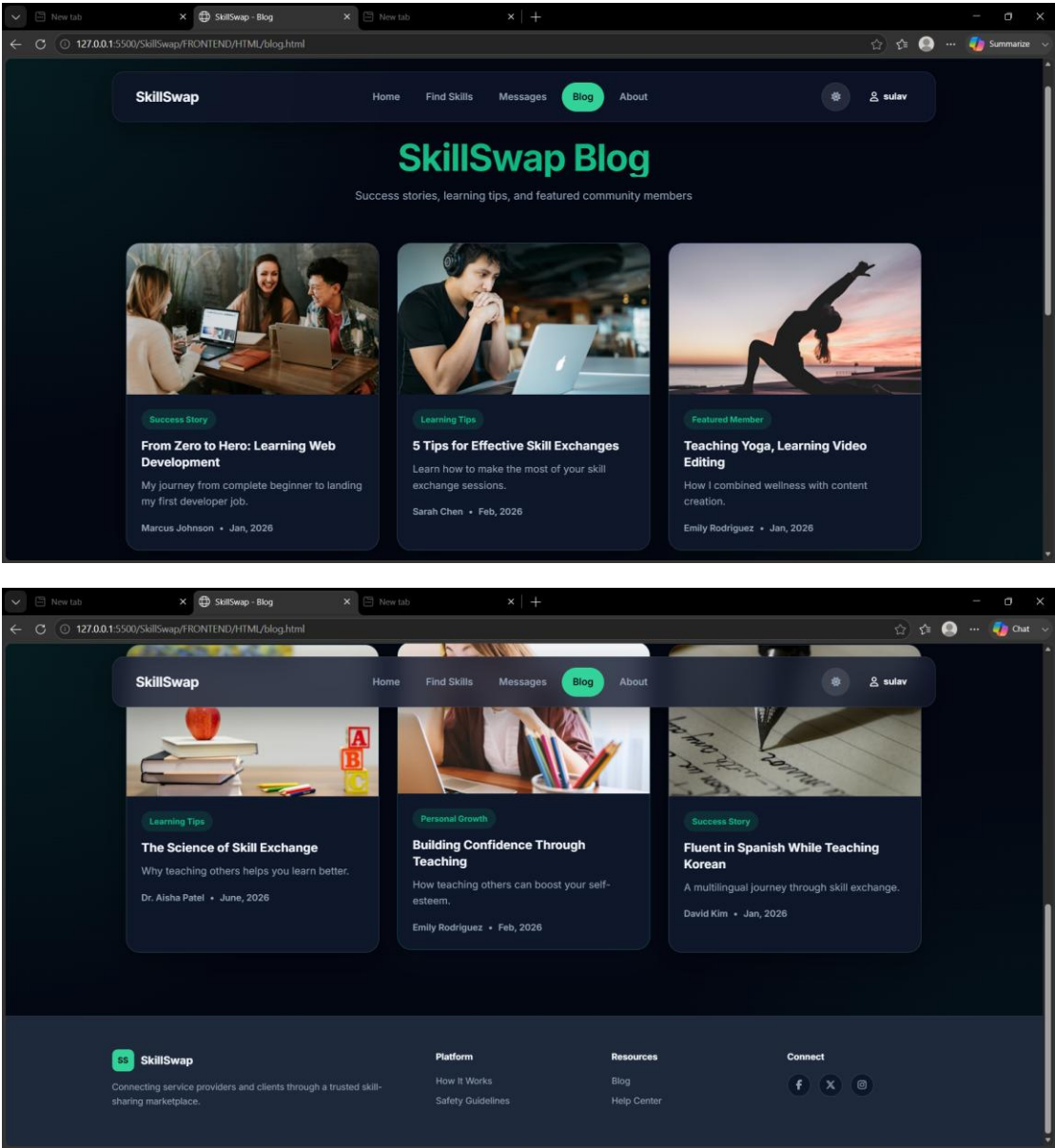




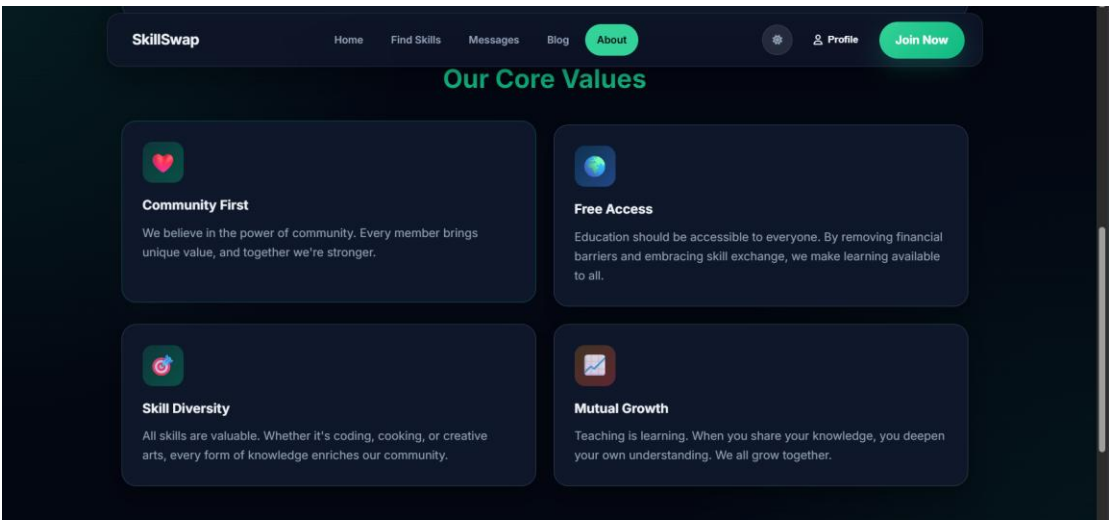
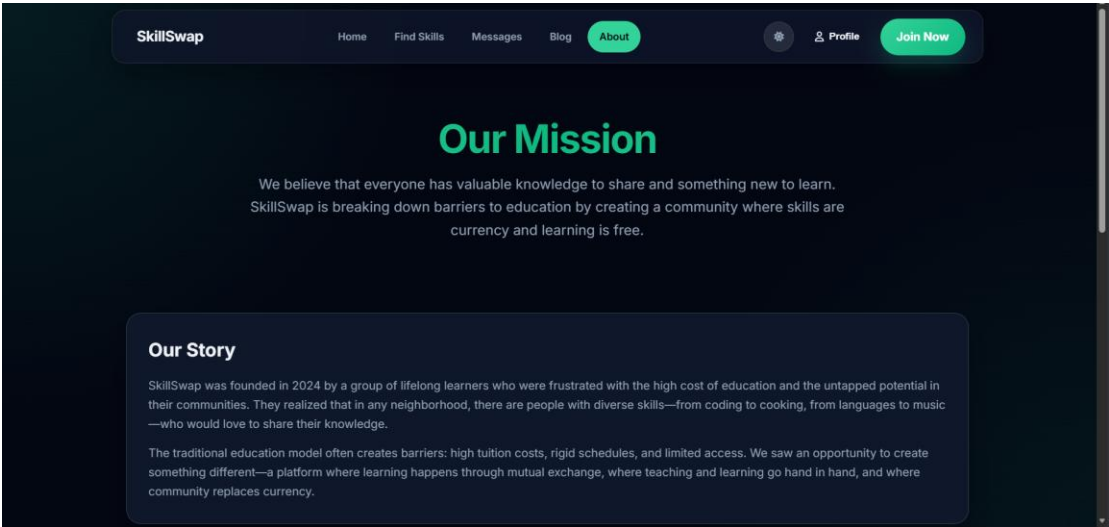
Find Skills Page Interface



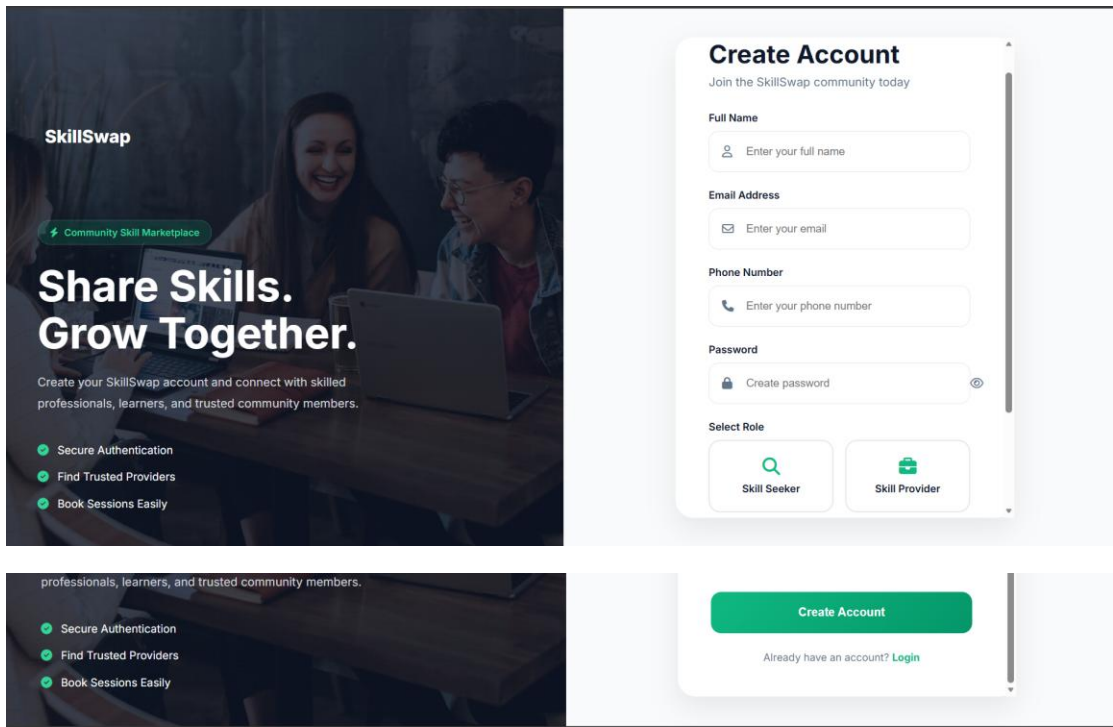
Blog Page Interface



About Page Interface



Sign Up Page Interface



The Sign Up Page Interface is divided into two main sections. The left section features a dark background with a photo of two people working on laptops. It includes the SkillSwap logo, a tagline 'Share Skills. Grow Together.', a sub-header 'Community Skill Marketplace', and a list of benefits: 'Secure Authentication', 'Find Trusted Providers', and 'Book Sessions Easily'. The right section is a white card titled 'Create Account' with the sub-header 'Join the SkillSwap community today'. It contains input fields for 'Full Name', 'Email Address', 'Phone Number', and 'Password'. Below these is a 'Select Role' section with two buttons: 'Skill Seeker' and 'Skill Provider'. At the bottom of the card is a green 'Create Account' button and a link 'Already have an account? Login'.

SkillSwap

Community Skill Marketplace

Share Skills. Grow Together.

Create your SkillSwap account and connect with skilled professionals, learners, and trusted community members.

- Secure Authentication
- Find Trusted Providers
- Book Sessions Easily

Create Account

Join the SkillSwap community today

Full Name

Email Address

Phone Number

Password

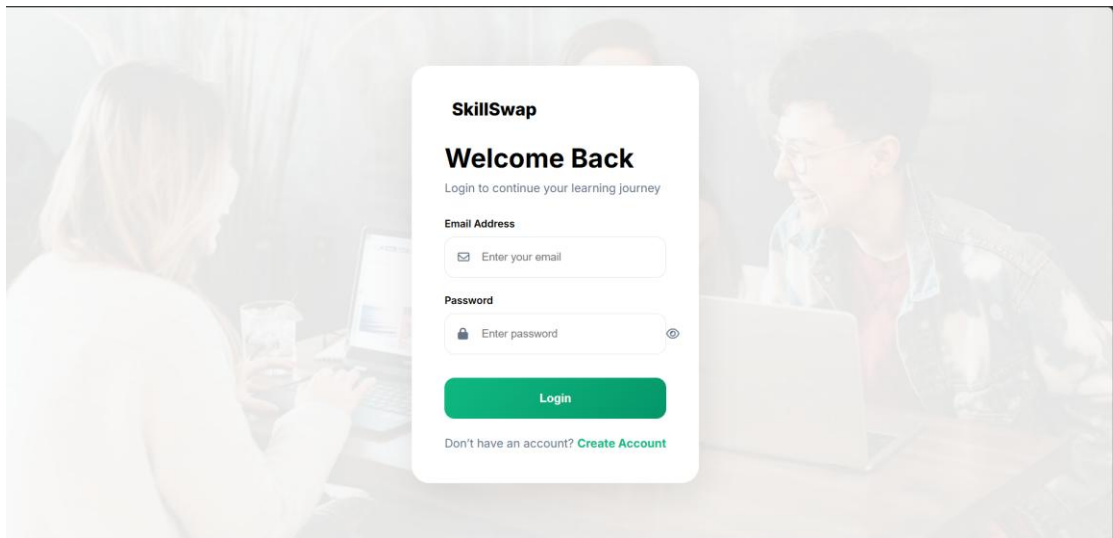
Select Role

[Skill Seeker](#) [Skill Provider](#)

[Create Account](#)

[Already have an account? Login](#)

Login Page Interface



The Login Page Interface features a light gray background with a faded image of two people working on laptops. A white card is centered on the page. The card has the SkillSwap logo, the title 'Welcome Back', and the sub-header 'Login to continue your learning journey'. It contains input fields for 'Email Address' and 'Password'. Below these is a green 'Login' button. At the bottom of the card is a link 'Don't have an account? Create Account'.

SkillSwap

Welcome Back

Login to continue your learning journey

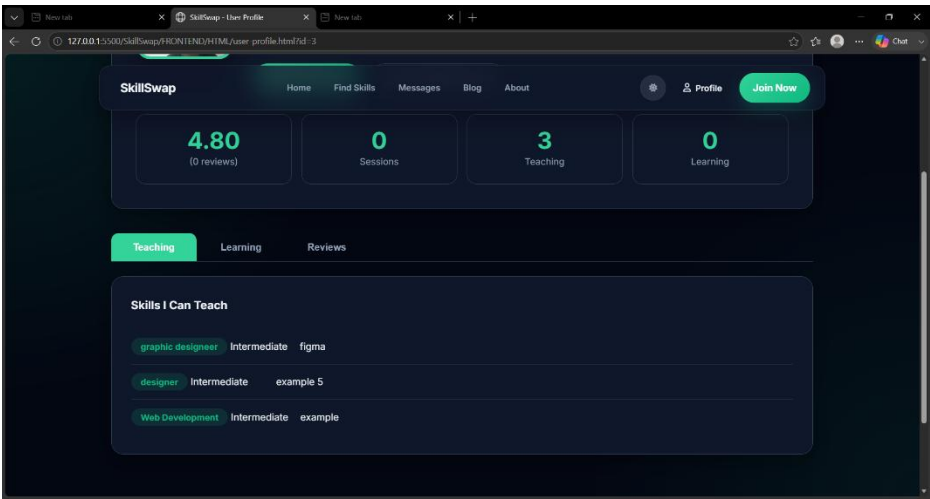
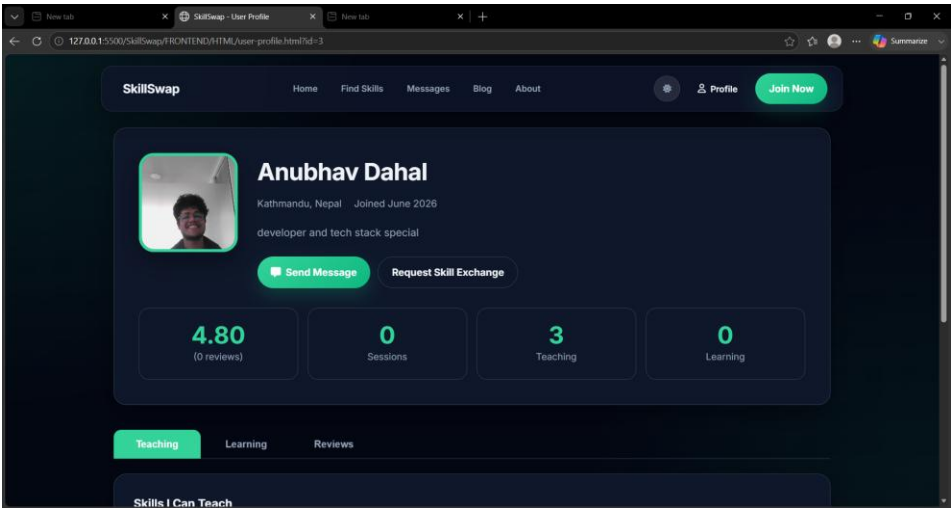
Email Address

Password

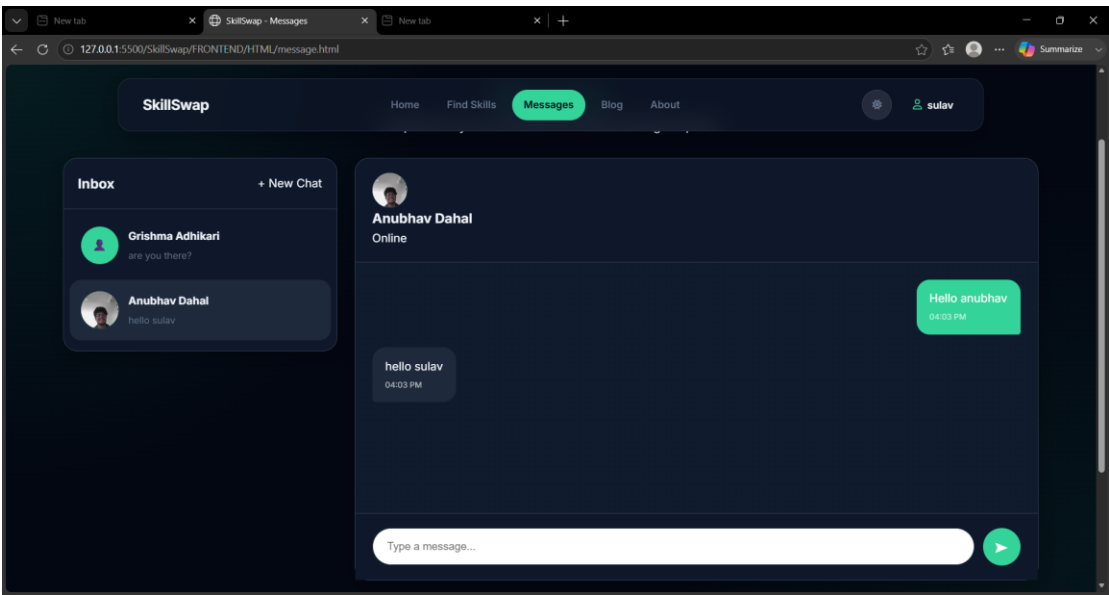
[Login](#)

Don't have an account? [Create Account](#)

View Profile Interface



Message Page Interface



REFERENCES

- [1] Sri, M. B. M. (2025). *Peer-to-peer skills and knowledge exchange platform*. International Journal of Scientific Research in Engineering and Management, 9(2).https://www.researchgate.net/publication/389074921_Peer-to-Peer_Skills_and_Knowledge_Exchange_Platform
- [2] Jadhav, A. D., et al. (2025). *SkillMesh: A peer-to-peer platform for collaborative skill exchange*. International Journal of Science and Research Archive, 15(1), 1544–1551. <https://ijsra.net/node/1024>
- [3] Anam, C., et al. (2025). *Design and implementation of SkillShare Campus: A web-based collaborative learning platform for higher education*. Journal of Applied Informatics Research, 1(2). <https://journal.unesa.ac.id/index.php/jair/article/view/49965>
- [4] Ukata, D. J., & Moko, A. B. (2026). *A mobile-based collaborative learning application for peer-to-peer SkillSwap among university students*. International Journal of Innovative Science and Research Technology. https://www.researchgate.net/publication/401320248_A_Mobile-Based_Collaborative_Learning_Application_for_Peer-to-Peer_SkillSwap_Among_University_Students